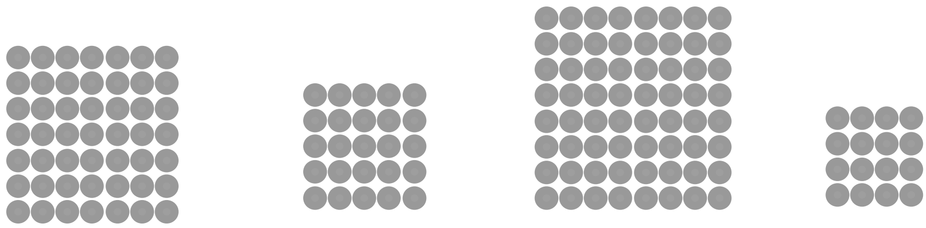
Name: ____________________

Worksheet

# Square and cube numbers

## Task A: Square numbers

1) Match up the arrays with the correct square number.



64

49

16

25

2) Work out the age given the clues.

My teacher's age is:

a square number,

a multiple of 3,

and less than 70.

How old is the teacher?

Alex is a student at Oak Academy.

Alex's age is:

4 less than a square number and is an even number.

How old is Alex?

3) Identify if the following statements are true or false. Justify your answer.

| Statement | True | False | Justification |
|---|---|---|---|
| 121 is a square number | | | |
| $13 \times 13 = 169$ so 13 is a square number | | | |
| 4 add 3 then multiply by 7 gives a square number | | | |
| 7 multiplied by 4 then add 3 gives a square number | | | |

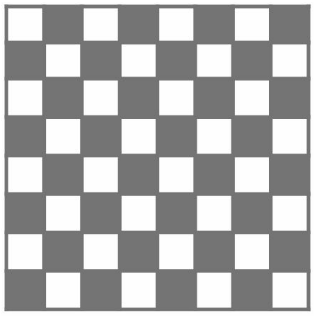
4) Jun discovers he can sum two different square numbers to make a third square number.

a) Can you find the missing square numbers to complete this table?

| 1st square number | 2nd square number | Resultant square number |
|---|---|---|
| 9 | 16 | 25 |
| | | 100 |
| | | 169 |

b) Can you find another set of two different square numbers to make a third square number?

5) How many squares can you see on an 8 by 8 chess board?



## Task B: Cube numbers

1) Identify all the cube numbers from the list below.

1 2 3 4 8 9 10

16 25 27 36 40 49

64 100 121 125 169

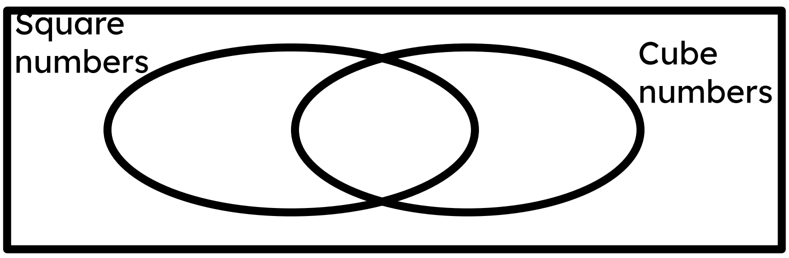
1000

2) Put the following numbers in the Venn diagram:

1 3 4 8 9

16 25 27 36 40 49

64 125 216 1000



3) Fill in the number to make the calculation correct.

The number in the SQUARE is a square number

The number in the CUBE is a cube number.

a) ⬜ + 🧊 = 31
b) ⬜ + 🧊 = 33
c) ⬜ + 🧊 = 225
d) ⬜ + ⬜ + 🧊 = 104
e) 🧊 + 🧊 + 🧊 = 36